

CS6522: PROJECT IN AUTONOMOUS DRIVING

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Project in Autonomous Driving

Subject Code

CS - Computer Science

Course Number

6522

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

Two Semesters

Credit Units

0-6

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Students should have completed at least 12 credit units (including two required courses) and achieve a GCPA ≥ 3.30 . Students with CGPA 3.00 - 3.29 may be considered on a case-by-case basis subject to departmental approval. Students with CGPA below 3.00 are not eligible.

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

CS6523 Internship in Autonomous Driving or
CS6524 Project in Generative AI or
CS6525 Internship in Generative AI or
CS6526 Project in Trustworthy AI or
CS6527 Internship in Trustworthy AI or

CS6528 Internship in Artificial Intelligence or
 CS6529 Project in Artificial Intelligence or
 CS6539 Internship in Autonomous Driving or
 CS6540 Internship in Generative AI or
 CS6541 Internship in Trustworthy AI or
 CS6542 Internship in Artificial Intelligence

Part II Course Details

Abstract

This course aims to provide an opportunity for students to explore individually an area of AI-enabled autonomous driving systems on their own choice. It allows students to develop their skill and knowledge further in the area of interest. It provides the context for students to demonstrate their ability to integrate specialized knowledge that they have acquired in other preceding and concurrent courses of study and apply them to solve an advanced problem with a working solution.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if DEC-A1 app.)		DEC-A2	DEC-A3
1	Identify a challenging problem related to AI-enabled autonomous driving systems, analyze the problem in detail; and propose innovative solutions by techniques in computing and AI.		x		
2	Provide a proof-of-concept for the solution by designing and developing a working system or application.			x	x
3	Implement and evaluate the developed system or application to match the initial system requirements.				x
4	Document and report the system design process, study, implementation and evaluation findings using different communication media.				x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Project planning	Students will identify the problem for investigation and draft a project plan with appropriate milestones.	1	

2	Project proposal	Students will analyze the problem identified and research on existing and/or related solutions. Then, in consultation with their supervisors, they will propose their own designs and solutions.	2	
3	Project implementation and evaluation	Students will implement the proposed solutions and validate their designs by testing and evaluating the completed solution.	3	
4	Project documentation	Students will document and explain their work in regular progress reports and a final report. At the end, they are required to present their projects in the form of oral presentation and demonstration.	4	

Additional Information for LTAs

Teaching pattern:

Suggested lecture/tutorial/laboratory mix: 8 hours individual consultation per semester.

The course is designed to guide students in proposing and managing their own projects. Each student will find an academic staff to supervise the project on a one to one basis.

The role of the supervisor is to closely monitor the project progress with project meetings regularly, in order to give advice to the student, to establish criteria for assessment, and to advise on possible solutions and potential problems at an early stage. In particular, the supervisor is expected to encourage the student to explore innovative approaches and alert the student to the possibility of alternative and novel solutions to problems encountered.

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Topic Selection	1	0	Allowed for writing polishing but the ideas must remain strictly the student's own.	Yes
2	Project Proposal	2	5	Allowed for writing polishing but the ideas must remain strictly the student's own.	Yes
3	1st Interim Report	3	5	Allowed for writing polishing but the ideas must remain strictly the student's own.	Yes

4	2nd Interim Report	3	5	Allowed for writing polishing but the ideas must remain strictly the student's own.	Yes
5	Final Report	4	60	Allowed for writing polishing but the ideas must remain strictly the student's own.	Yes
6	Presentation	4	20	Allowed for writing polishing but the ideas must remain strictly the student's own.	Yes
7	Project Management	1	5	Allowed for writing polishing but the ideas must remain strictly the student's own.	Yes

Continuous Assessment (%)

100

Additional Information for ATs

For assessment of technical merit, report, and presentation, the project committee assigns two examiners, including the supervisor. The Supervisor is required to give detailed grading reports on all aspects of assessment. The Assessor will evaluate the CILOs 2-4 of the project. The Course Leader will review all projects, moderate consistency across a wide range of projects, and, where necessary, resolve discrepancies between grading of the Assessor and the Supervisor, drawing on the expertise of domain experts as needed.

Dissertation-type Course:

This course falls under the academic regulation for dissertation-type courses (AR12.6). The course assessed through 100% coursework.

Each student is assigned a supervisor from the academic staff for individual consultation.

The normal duration of the course is two semesters, after which the dissertation must be submitted.

The maximum duration of the course is four semesters, after which no further extension must be permitted.

Dissertation-type courses may NOT be repeated.

Assessment Rubrics (AR)**Assessment Task**

Topic Selection & Project Management

Criterion

1.1 ABILITY to IDENTIFY problems for investigations.

1.2 ABILITY to PLAN a project schedule with appropriate milestones, and MAINTAIN the project schedule.

Excellent (A+, A, A-)

High

Good (B+, B, B-)

Significant

Fair (C+, C, C-)

Moderate

Marginal (D)

Basic

Failure (F)

Not even reaching marginal levels

Assessment Task

Project Proposal

Criterion

2.1 ABILITY to ANALYZE a problem.

2.2 ABILITY to EVALUATE, COMPARE, and CONTRAST existing solutions.

2.3 ABILITY to DESIGN and INNOVATE new solutions.

Excellent (A+, A, A-)

High

Good (B+, B, B-)

Significant

Fair (C+, C, C-)

Moderate

Marginal (D)

Basic

Failure (F)

Not even reaching marginal levels

Assessment Task

1st Interim Report & 2nd Interim Report

Criterion

3.1 ABILITY to IMPLEMENT the proposed solution.

3.2 ABILITY to VALIDATE and TEST the implemented solution.

3.3 ABILITY to EVALUATE and INTERPRET results from the design, and COMPARE with existing solutions.

Excellent (A+, A, A-)

High

Good (B+, B, B-)

Significant

Fair (C+, C, C-)

Moderate

Marginal (D)

Basic

Failure (F)

Not even reaching marginal levels

Assessment Task

Final Report & Presentation

Criterion

4.1 ABILITY to DOCUMENT the progress of the project in interim reports.

4.2 ABILITY to DOCUMENT the OUTCOMES of the project in a final report.

4.3 ABILITY to DEMONSTRATE project outcomes in an oral presentation.

Excellent (A+, A, A-)

High

Good (B+, B, B-)

Significant

Fair (C+, C, C-)

Moderate

Marginal (D)

Basic

Failure (F)Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

The Project has no fixed formal syllabus. Each student will be required to undertake an individual piece of work, which is related to the area of AI-enabled autonomous driving systems. The topic area of the dissertation will be chosen so that the aims of the Project can be achieved. Criteria for topic choice include: (i) compatibility with a subject area of AI-enabled autonomous driving systems, (ii) availability of a qualified supervisor; (iii) appropriate academic level; (iv) availability of necessary specialized resources. Topic areas include: sensing, perception, tracking, localization, prediction, planning, driving control, system software and middleware for autonomous driving systems, performance optimization of autonomous driving systems, real-time performance analysis, safety and security of autonomous driving systems, hardware platform for autonomous driving systems.

Reading List**Compulsory Readings**

Title	
1	N/A

Additional Readings

Title	
1	N/A